








Time Series Analysis

Chapter 7: Cointegration and VECM



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Acronyms used in this material

- **ACF**: Autocorrelation Function
- **ADF**: Augmented Dickey–Fuller (test)
- **AI**: Artificial Intelligence
- **AIC**: Akaike Information Criterion
- **AR**: AutoRegressive (model)
- **ARCH**: AutoRegressive Conditional Heteroskedasticity
- **ARFIMA**: AutoRegressive Fractionally Integrated Moving Average (model)
- **BIC**: Bayesian Information Criterion
- **BNR**: Banca Națională a României (the National Bank of Romania)
- **BVB**: Bursa de Valori București (the Bucharest Stock Exchange)
- **DHSY**: Davidson, Hendry, Srba and Yeo (1978)
- **DOLS**: Dynamic Ordinary Least Squares
- **DW**: Durbin–Watson (statistic)
- **ECM**: Error Correction Model
- **FRED**: Federal Reserve Economic Data
- **GDP**: Gross Domestic Product
- **HICP**: Harmonised Index of Consumer Prices
- **LR**: Likelihood Ratio (test)
- **OLS**: Ordinary Least Squares
- **PPP**: Purchasing Power Parity
- **RMSE**: Root Mean Squared Error
- **ROBOR**: Romanian Interbank Offer Rate
- **SE**: Standard Error
- **UK**: United Kingdom
- **US**: United States
- **VAR**: Vector AutoRegression
- **VECM**: Vector Error Correction Model

Today's question and route

- **Question**: two prices both wander like random walks; can a combination of them be stable, and how do we model it?
  - if yes, the series share a common stochastic trend and a long-run equilibrium: regressions in levels are meaningful, not spurious
- **Route** of the chapter
  - the idea of cointegration; the Engle–Granger two-step method with the right critical values; error correction models
  - the VECM and the Granger representation theorem; the Johansen tests; estimating and reading β and α; forecasting
  - applications: purchasing power parity, interest rates, three Central European currencies, pairs trading on the Bucharest Stock Exchange
- We build on Chapter 3 (unit roots, spurious regression) and Chapter 6 (VAR models)

Learning outcomes

- Define cointegration and explain it as a common stochastic trend
- Run the Engle–Granger and Phillips–Ouliaris tests and compare them with the right (not the Dickey–Fuller) critical values
- Estimate an error correction model and interpret the speed of adjustment and the half-life
- Write a VAR as a VECM, state the Granger representation theorem and choose the cointegration rank with the Johansen tests
- Interpret cointegrating vectors and adjustment coefficients, and decide when a VECM forecasts better than a VAR in differences
- Judge an application (parity conditions, pairs trading) out of sample and net of costs

Reading and tools

- Textbook: Huang and Petukhina (2022), Ch. 9 (nonstationarity and cointegration)
  - theory: Hamilton (1994), Ch. 19–20; Lütkepohl (2005), Ch. 6–8; the likelihood approach in Johansen (1995) and Juselius (2006)
- Landmark papers: Engle and Granger (1987), Johansen (1988), Johansen (1991); Granger's Nobel lecture Granger (2004)
- Python Quantlets of this chapter: Quantlets/Ch_07
  - `coint`, `coint_johansen`, `VECM` and `VAR` from `statsmodels`; `phillips_ouliaris` from `arch`
- Lecture notebook: open in Google Colab
- Video course: Applied Time Series Analysis with Python

From spurious regression to cointegration

- Chapter 3: regressing one $I(1)$ series on another, **unrelated** $I(1)$ series gives large t, high R^2 and a Durbin–Watson statistic near 0 Granger and Newbold (1974)
  - the error $y_t - a - bx_t$ is itself $I(1)$ for every b: nothing ties the two series together
- The exception: for one value of b, the combination $u_t = y_t - a - bx_t$ is **stationary**
  - then y_t and x_t are **cointegrated** Granger (1981), Engle and Granger (1987): the regression in levels estimates a real long-run relation
  - the two series may drift apart for a while, but u_t always pulls them back: an **equilibrium**
- Today: how to recognise this case, how to test it, and how to model it

A drunk and her dog

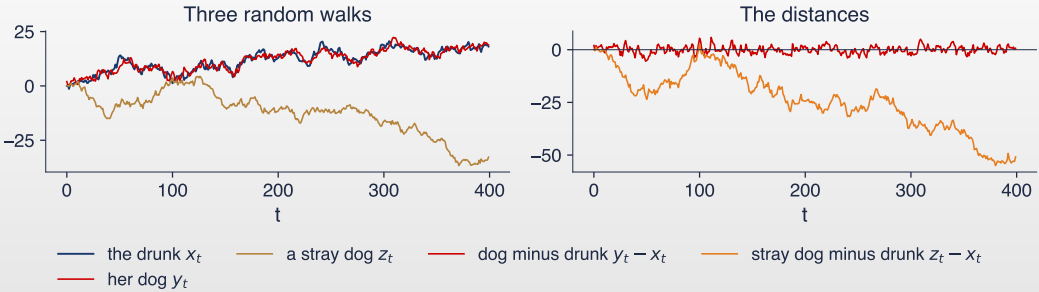


A walk with a dog: two paths that never drift far apart

Photo: Amin (2018); CC BY-SA 4.0; Wikimedia Commons

- Murray (1994): a drunk leaves the pub and wanders: her path x_t is a random walk
  - her dog also wanders: y_t is a random walk too
  - the distance between two **independent** walkers grows without limit, like $\sqrt{t}$
- But the dog hears her call and runs back; she hears the dog and turns towards it
  - each corrects part of the gap: $\Delta x_t = c\,(y_{t-1} - x_{t-1}) + u_t$, $\Delta y_t = d\,(x_{t-1} - y_{t-1}) + w_t$; $c, d > 0$: the shares of the gap closed per step; u_t, w_t: independent shocks
  - both paths remain unpredictable, but the distance $y_t - x_t$ stays bounded: **cointegration** with **error correction**

The drunk and her dog by simulation



- $T = 400$, Gaussian shocks; $c = 0.15$, $d = 0.25$; a stray dog z_t is an independent random walk with the same shocks' variance

TSA_ch7_common_trends

Interpreting the simulation: the distances

- Each path is $I(1)$: ADF on x_t gives p $= 0.21$
  - the distance $y_t - x_t$ is an AR(1) with $\rho = 1 - c - d = 0.6$: ADF p < 0.001, standard deviation 1.9
  - the distance to the stray dog wanders: ADF p $= 0.81$, standard deviation 14.1
- Half of a gap disappears in $\ln 0.5 / \ln 0.6 = 1.4$ steps: the **half-life** (Section 3)
  - knowing where the drunk is tells us where to look for the dog, even though neither path can be forecast

Definition of cointegration

- **Definition** Engle and Granger (1987): the components of $\mathbf{y}_t = (y_{1t}, \ldots, y_{nt})^\top$ are **cointegrated of order (1,1)**, $\mathbf{y}_t \sim CI(1,1)$, if
  - every component is $I(1)$ (Chapter 3: stationary after one difference), and
  - some linear combination $\beta^\top \mathbf{y}_t$, with $\beta \neq 0$, is $I(0)$: stationary
- β is the **cointegrating vector**; $u_t = \beta^\top \mathbf{y}_t - \mu$ is the **equilibrium error** (μ: its mean; $\top$: transpose)
  - β is unique only up to scale: 2β works as well; we **normalise** one coefficient to 1, e.g. $\beta = (1, -b)^\top$
  - with n variables there can be up to $r = n - 1$ independent cointegrating vectors; r is the **cointegration rank**
- Cointegration is a property of the **levels**: differencing the data destroys the information in $\beta^\top \mathbf{y}_t$

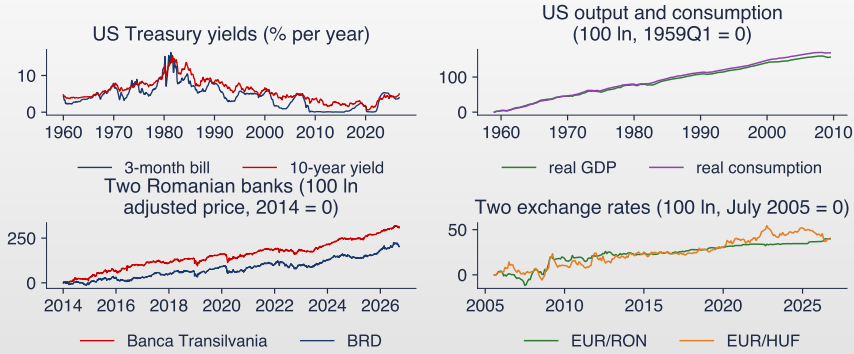
Common stochastic trends

- Write $x_t = x_{t-1} + \varepsilon_t$ (a random walk) and $y_t = b\,x_t + u_t$, with u_t stationary
  - both series contain the same stochastic trend $\sum_{s \le t} \varepsilon_s$; $y_t - b x_t = u_t$ removes it
- **Common trends** representation Stock and Watson (1988): n variables with r cointegrating vectors are driven by $n - r$ common stochastic trends
  - $r = 0$: n separate trends, no cointegration; $r = n - 1$: one trend shared by all
  - example: three yields (1, 5 and 10 years) with $r = 2$ share one trend, the **level** of interest rates (Section 5)
- **Worked example**: x_t a random walk, $y_t = 2x_t + u_t$, $z_t = -x_t + v_t$ (u_t, v_t stationary)
  - $n = 3$, one common trend, so $r = 2$: in the order (x_t, y_t, z_t), e.g. $\beta_1 = (-2, 1, 0)^\top$ and $\beta_2 = (0, 1, 2)^\top$ (since $y_t - 2x_t = u_t$ and $y_t + 2z_t = u_t + 2v_t$)

Where economics predicts cointegration

- **Purchasing power parity**: $s_t = p_t - p_t^* + q_t$, with the real exchange rate q_t stationary (Section 8)
  - s_t: log exchange rate (lei per euro); p_t, p_t^*: log price levels at home and abroad
- **Term structure of interest rates**: by the expectations hypothesis, the spread between a long and a short yield is stationary Campbell and Shiller (1987)
- **Consumption and income**: the "great ratio" $c_t - y_t$ (logs) is stable if consumption follows permanent income
- **Spot and futures prices**, an index and a fund that tracks it: arbitrage keeps the gap small
- **Shares of similar firms**: the same sector shocks; the basis of pairs trading (Section 9)

Four pairs of trending series



- US Treasury yields (FRED, monthly, since 1960); US real GDP and consumption (statsmodels macrodata, 1959–2009)
- Banca Transilvania and BRD (adjusted daily prices); EUR/RON and EUR/HUF (BNR reference rates, month-end)

TSA_ch7_common_trends

Interpreting the four pairs

- Each series is $I(1)$ (Chapter 3); the question is whether the two series of a pair move **together** in the long run
  - Engle–Granger test (Section 2), 5% critical value about -3.34
- Yields: $\tau = -3.35$, p $= 0.048$: borderline; consumption and income: $\tau = -3.54$, p $= 0.029$
- The two banks: $\tau = -4.33$, p $= 0.002$: cointegrated in this sample
- EUR/RON and EUR/HUF: $\tau = -2.47$, p $= 0.294$: **no evidence** of cointegration
  - although they rose by similar amounts and $R^2 = 0.79$ in levels: a common direction is not a common trend

Question for the room

- **Question**: two $I(1)$ series have a correlation of 0.95 in levels. Are they cointegrated?
  - **Answer**: we cannot tell: unrelated random walks often have high correlations (Chapter 3); only a test on the equilibrium error decides
- **Question**: two cointegrated series have a correlation of 0.2 between their daily changes. Is this a contradiction?
  - **Answer**: no: cointegration is about the long run; in the short run the changes can be almost unrelated

Recap: The idea of cointegration

- $I(1)$ series are cointegrated if a combination $\beta^\top \mathbf{y}_t$ is stationary: a long-run equilibrium
- Equivalently, they share common stochastic trends: n variables, r vectors, $n - r$ trends
- Theory suggests cointegration (parity conditions, spreads, great ratios), but the data must confirm it

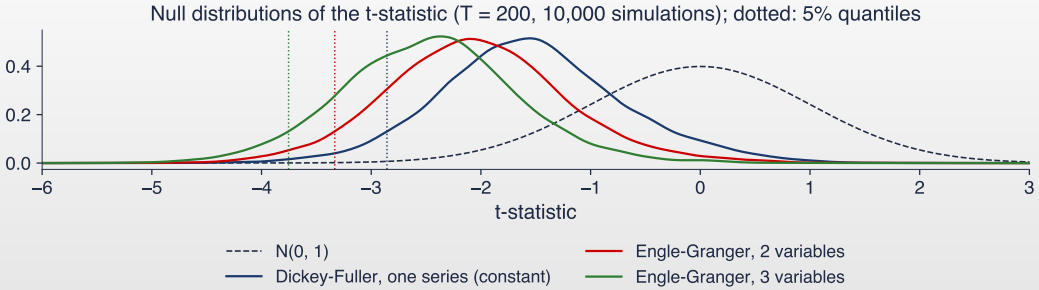
Two steps

- **Step 1**: estimate the long-run relation by OLS (ordinary least squares) in levels: $y_t = a + b\,x_t + u_t$
  - keep the residuals $\hat{u}_t = y_t - \hat{a} - \hat{b}x_t$
  - if the series are cointegrated, $\hat{b}$ is **super-consistent**: its error shrinks like $1/T$, faster than the usual $1/\sqrt{T}$
- **Step 2**: an ADF test (Chapter 3) on the residuals, without a constant: $\Delta\hat{u}_t = \gamma\,\hat{u}_{t-1} + \sum_{j=1}^{k}\delta_j\Delta\hat{u}_{t-j} + e_t$
  - H_0: $\gamma = 0$, the residuals have a unit root: **no cointegration**; H_1: $\gamma < 0$: cointegration
  - δ_j: coefficients of the k lagged differences, which absorb the autocorrelation of $\hat{u}_t$; e_t: white noise
  - statistic $\tau = \hat{\gamma}/\mathrm{SE}(\hat{\gamma})$, SE: standard error; reject H_0 when τ is below the critical value
- Pretest: each series must be $I(1)$; cointegration between an $I(0)$ and an $I(1)$ series is impossible

The reason the Dickey–Fuller table is wrong here

- OLS chooses $\hat{b}$ to make the residual variance as small as possible
  - so $\hat{u}_t$ looks more stationary than the true u_t, even when there is no cointegration
  - under H_0 the statistic τ is shifted to the left of the Dickey–Fuller distribution
- The critical values depend on the number of variables n and on the deterministic terms of step 1
  - tables: Engle and Granger (1987) (by simulation), then MacKinnon (1994) and MacKinnon (2010), used by `statsmodels` (`coint`)
- With β **known** from theory (e.g. a spread, $\beta = (1, -1)^\top$), nothing is estimated: the plain ADF test and its table apply

The null distribution by simulation



- 10 000 samples of $T = 200$ independent Gaussian random walks; step 1 with a constant, step 2 without lags

TSA_ch7_engle_granger

Interpreting the simulation: the critical values

- 5% quantiles: -2.86 (one series, Dickey–Fuller), -3.33 (2 variables), -3.75 (3 variables)
  - MacKinnon's values for $T = 200$: -2.88; -3.37; -3.78: the simulation reproduces them
- Using the Dickey–Fuller value -2.88 for the residuals would reject a true H_0 in 14.5% of samples with 2 variables and 30.3% with 3
  - instead of 5%: we would "find" cointegration far too often
- More variables give OLS more freedom to fit, hence more negative critical values

Critical values of the Engle–Granger test

Variables n	constant			constant and trend		
	1%	5%	10%	1%	5%	10%
1 (Dickey–Fuller)	−3.43	−2.86	−2.57	−3.96	−3.41	−3.13
2	−3.90	−3.34	−3.04	−4.33	−3.78	−3.50
3	−4.29	−3.74	−3.45	−4.66	−4.12	−3.84
4	−4.64	−4.10	−3.81	−4.97	−4.43	−4.15

- Asymptotic values ($T \to \infty$) of MacKinnon (2010); for finite T the values are slightly more negative
- Read the row of the **total** number of variables in the regression, y_t included
- Constant and trend in step 1: when the series have drifts that the relation does not cancel

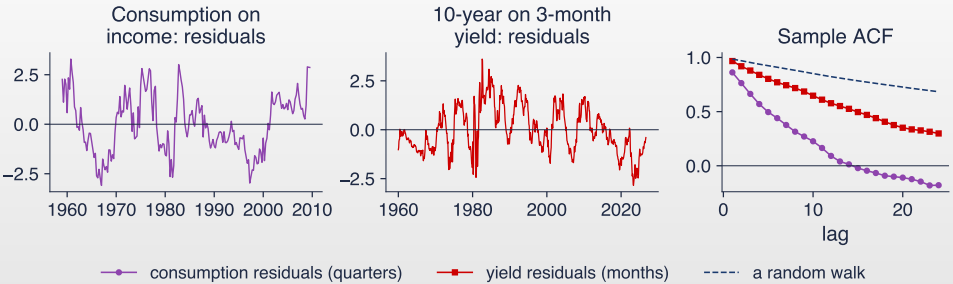
The Phillips–Ouliaris test

- Phillips and Ouliaris (1990): the same step 1 and the same H_0 (no cointegration)
  - step 2: a Dickey–Fuller regression **without** lagged differences; the t-statistic is corrected with the long-run variance of the residuals (as in the Phillips–Perron test of Chapter 3)
  - two versions: Z_t (corrected t-ratio) and Z_α (corrected $T\hat{\gamma}$); the critical values of Z_t are those of Engle–Granger
- Reasons for two tests
  - Engle–Granger depends on the number of lags k chosen by AIC; Phillips–Ouliaris depends on the bandwidth of the long-run variance
  - when they agree, the verdict is robust; when they disagree, look at the residual plot and at the sample
- In Python: `coint(y, x)` from `statsmodels`; `phillips_ouliaris(y, x, test_type="Zt")` from `arch`

Worked example: US consumption and income

- Data: c_t, $y_t = 100 \times$ log of real consumption and real GDP, quarterly, 1959Q1–2009Q3 (203 quarters)
  - pretest: ADF with trend, p = 0.31 for c_t and 0.39 for y_t: both $I(1)$
- Step 1: $\hat{c}_t = -107.6 + 1.075\, y_t$, $R^2 = 0.9992$, DW $= 0.24$
  - DW: the Durbin–Watson statistic of the residuals, about 2 without autocorrelation and close to 0 for very persistent residuals
  - the slope is close to 1: a stable consumption share; but R^2 and DW alone prove nothing
- Step 2: $\tau = -3.54$ with 0 lags; 5% critical value for $n = 2$, $T = 203$: -3.37
  - $-3.54 < -3.37$: reject "no cointegration" at 5% (p = 0.029); Phillips–Ouliaris: $Z_t = -3.52$, p = 0.031
  - with the Dickey–Fuller value (-2.88) the decision would be the same here, but for the wrong reason

The step-1 residuals



- Left and centre: $\hat{u}_t$ of consumption on income and of the 10-year on the 3-month Treasury yield; right: their sample ACF and that of a simulated random walk

TSA_ch7_engle_granger

Interpreting the residuals

- Consumption: $\hat{\rho}(1) = 0.86$ but $\hat{\rho}(8) = 0.31$: deviations of 1–3% that shrink markedly within about two years
- Yields: $\hat{\rho}(1) = 0.97$, $\hat{\rho}(12) = 0.58$: very persistent deviations; a borderline test ($\tau = -3.35$, p = 0.048)
  - high persistence is the usual situation: the power of the test against ρ close to 1 is low (Chapter 3)
- A random walk decays much more slowly: the ACF alone cannot settle the question; the test can

Engle–Granger and Phillips–Ouliaris on real data

Regression	Freq.	T	$\hat{b}$	τ (k)	p	Z_t	p
US consumption on income	quarterly	203	1.07	−3.54 (0)	0.029	−3.52	0.031
US: 10-year on 3-month yield	monthly	801	0.86	−3.35 (20)	0.048	−4.04	0.006
TLV on BRD (since 2014)	daily	3 176	1.36	−4.33 (3)	0.002	−4.14	0.005
BRD on TLV (since 2014)	daily	3 176	0.71	−4.16 (3)	0.004	−4.02	0.007
TLV on BRD (since 2010)	daily	4 187	1.68	−1.82 (3)	0.622	−1.83	0.619
EUR/RON on EUR/HUF	monthly	255	0.71	−2.47 (0)	0.294	−2.43	0.309
Romania: consumption on GDP	quarterly	126	1.46	−2.32 (8)	0.363	−6.80	< 0.001

- τ: Engle–Granger statistic (k lags by AIC); p-values of MacKinnon (2010) for 2 variables; Z_t: Phillips–Ouliaris
- Banks: adjusted prices from data/market; Romania: household consumption and GDP, Eurostat, chain-linked volumes, seasonally adjusted, since 1995

Interpreting the table

- **Normalisation**: TLV on BRD and BRD on TLV give different τ (-4.33 and -4.16) and slopes that are not exact inverses
  - Engle–Granger depends on which variable is on the left; here the verdict is the same
- **The sample**: since 2010 the test finds no cointegration for the same banks ($p = 0.622$); since 2014 it does ($p = 0.002$)
  - choosing the start date after looking at the data is a form of data snooping
- **The two tests disagree** for Romania: Engle–Granger $p = 0.363$ with 8 lags, Phillips–Ouliaris $p < 0.001$
  - the slope 1.46 is far from 1: the consumption share rose strongly; Seminar 7 (B2) discusses this case

Limits of the Engle–Granger method

- **At most one** cointegrating vector: with $n \geq 3$ variables there may be several (Sections 4–5)
- **Normalisation**: the results depend on the variable put on the left
- **Inference on** $\hat{b}$: super-consistent, but biased in small samples, and its OLS t-statistics are **not** valid
  - remedy: dynamic OLS (DOLS) Stock and Watson (1993), adding leads and lags of Δx_t, gives valid standard errors
- **Two-step errors**: mistakes in step 1 carry over to step 2; the power is low against slow adjustment
  - the Johansen method (Section 5) estimates everything at once, by maximum likelihood

Case study: Engle and Granger (1987)



Robert Engle (2022); Nobel Prize in Economics 2003

Photo: Jon Demske (2022); CC BY-SA 4.0; Wikimedia Commons

- **Question**: how to model $I(1)$ series that are tied in the long run, without spurious regressions and without differencing away the long run?
  - Engle and Granger (1987), written at the University of California, San Diego, one of the most cited papers in econometrics
- **Contribution**
  - the definition of cointegration and the representation theorem (Section 4): cointegration $\Leftrightarrow$ error correction
  - the two-step estimator and residual-based tests with simulated critical values
  - applications to US series such as consumption and income, prices and wages, short and long interest rates

Recap: The Engle–Granger method

- Step 1: OLS in levels (super-consistent $\hat{b}$); step 2: ADF on the residuals, H_0: no cointegration
- Critical values of Engle–Granger (MacKinnon), more negative than Dickey–Fuller and dependent on n
- Check with Phillips–Ouliaris, with the reverse normalisation and with another sample

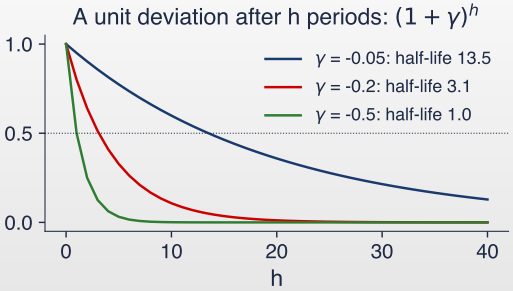
The error correction model

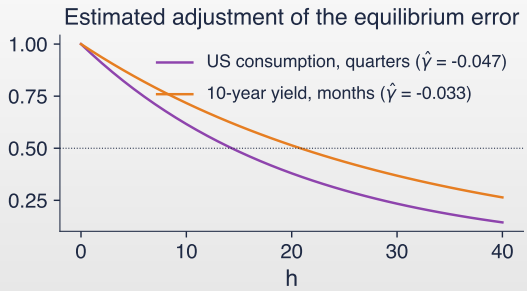
- With $u_{t-1} = y_{t-1} - a - bx_{t-1}$ (last period's equilibrium error), the **error correction model** (ECM) is
  - $\Delta y_t = c + \gamma\, u_{t-1} + \delta_0\, \Delta x_t + \sum_{j\geq 1}(\phi_j \Delta y_{t-j} + \delta_j \Delta x_{t-j}) + \varepsilon_t$
- Every term is stationary: OLS and the usual t-tests are valid (with $\hat{u}_{t-1}$ from step 1)
  - b: the **long-run** effect of x on y; δ_0: the **short-run** (same-period) effect
  - γ: the **speed of adjustment**; error correction requires $-2 < \gamma < 0$
  - c: constant; ϕ_j, δ_j: short-run coefficients of the lagged differences; ε_t: white noise
- Reading: if y_{t-1} was above equilibrium ($u_{t-1} > 0$) and $\gamma < 0$, then y_t tends to fall: a share $|\gamma|$ of the gap is closed each period

Speed of adjustment and half-life

- Without new shocks, the equilibrium error evolves as $u_t = (1+\gamma)\, u_{t-1}$, so $u_h = (1+\gamma)^h u_0$
  - (in a bivariate system where only y adjusts)
- **Half-life**: the number of periods after which half of a deviation is gone
  - $(1+\gamma)^h = 0.5 \;\Rightarrow\; h_{1/2} = \ln 0.5 / \ln(1+\gamma)$
- **Worked example**: $\gamma = -0.2$, $h_{1/2} = \ln 0.5 / \ln 0.8 = 3.1$ periods
  - in months: about three months for half of a deviation; in quarters: three quarters
- Compare half-lives only in the same time unit: a monthly and a quarterly γ are not comparable

Half-lives: theory and estimates





⊡ Left: $(1+\gamma)^h$ for three speeds; right: the decay implied by the estimated ECMs of US consumption (quarters) and of the 10-year yield (months)

TSA_ch7_error_correction

Interpreting the estimated ECMs

- Consumption: $\hat{\gamma} = -0.047$ ($t = -1.75$), half-life 14.3 quarters; $\hat{\delta}_0 = 0.54$
  - half of a rise in income reaches consumption in the same quarter; the rest comes slowly, and the adjustment term is only marginally significant
- 10-year yield: $\hat{\gamma} = -0.033$ ($t = -4.94$), half-life 20.8 months; $\hat{\delta}_0 = 0.38$
  - the 3-month rate also reacts to the same gap: $\hat{\gamma} = 0.024$ ($t = 2.08$), with the opposite sign: both rates move towards each other
- When both variables adjust, a single-equation ECM tells only half of the story: hence the VECM (Section 4)

Worked example: one quarter of consumption

- Estimated ECM (main terms): $\Delta c_t = 0.418 + 0.54\,\Delta y_t - 0.047\,\hat{u}_{t-1} + \ldots$, with $\hat{u}_{t-1} = c_{t-1} - 1.075\,y_{t-1} - \hat{a}$
- Scenario: consumption is 2% above its equilibrium ($\hat{u}_{t-1} = 2$) and income grows by 1% this quarter
  - short-run effect: $0.537 \times 1 = 0.537$; error correction: $-0.047 \times 2 = -0.095$
  - predicted growth of consumption: about 0.86% (the lagged differences are set to 0)
- Without the error correction term, the model would ignore that consumption is already too high

Case study: the consumption function of DHSY (1978)

- **Question**: why did the consumption functions of the 1970s, estimated in levels or in differences, forecast UK consumers' expenditure so badly?
  - Davidson et al. (1978) (Davidson, Hendry, Srba and Yeo, "DHSY"): quarterly UK data on expenditure and income
- **Method**: a model in differences plus the lagged log ratio of expenditure to income, the error correction term
  - a general-to-specific search, starting from a large dynamic model
- **Legacy**: the ECM form came first; Granger (1981) and Engle and Granger (1987) then showed why it works: the two series are cointegrated
  - a test of cointegration can also use the t-statistic of γ in the ECM, with its own critical values Banerjee, Dolado and Mestre (1998)

Recap: Error correction models

- ECM: Δy_t on Δx_t, lagged differences and $\hat{u}_{t-1}$; all terms stationary
- $\gamma < 0$: speed of adjustment; half-life $\ln 0.5 / \ln(1+\gamma)$, in the units of the data
- Short run (δ_0) and long run (b) in one equation; if several variables adjust, we need a system

From a VAR to a VECM

- Chapter 6: a VAR(p) in levels, $\mathbf{y}_t = \mathbf{c} + A_1\mathbf{y}_{t-1} + \cdots + A_p\mathbf{y}_{t-p} + \mathbf{u}_t$
  - subtract $\mathbf{y}_{t-1}$ and regroup the lags: the same model in **vector error correction** form (VECM)
- $\Delta\mathbf{y}_t = \mathbf{c} + \Pi\,\mathbf{y}_{t-1} + \sum_{i=1}^{p-1}\Gamma_i\,\Delta\mathbf{y}_{t-i} + \mathbf{u}_t, \quad \Pi = A_1 + \cdots + A_p - I, \quad \Gamma_i = -(A_{i+1} + \cdots + A_p)$
  - Π: the long-run matrix (it multiplies the levels); Γ_i: the short-run matrices; I: the identity matrix
- **Derivation** for $p = 2$: $\mathbf{y}_t - \mathbf{y}_{t-1} = (A_1 - I)\mathbf{y}_{t-1} + A_2\mathbf{y}_{t-2} + \mathbf{u}_t$; add and subtract $A_2\mathbf{y}_{t-1}$:
  - $\Delta\mathbf{y}_t = (A_1 + A_2 - I)\,\mathbf{y}_{t-1} - A_2\,\Delta\mathbf{y}_{t-1} + \mathbf{u}_t$: $\Pi = A_1 + A_2 - I$, $\Gamma_1 = -A_2$
- The VECM is a rewriting, not a new model: all the information about the long run sits in the matrix Π

The rank of Π

- $\Delta \mathbf{y}_t$ is stationary; so $\Pi \mathbf{y}_{t-1}$ must be stationary too. Three cases, by $r = \text{rank}(\Pi)$:
- $r = 0$: $\Pi = 0$; no cointegration; the right model is a **VAR in differences**
- $r = n$: Π has full rank; $\mathbf{y}_t$ is already stationary; a **VAR in levels** (Chapter 6)
- $0 < r < n$: **cointegration** with r vectors; $\Pi = \alpha\beta^\top$ with α, β of size $n \times r$
  - $\beta^\top \mathbf{y}_{t-1}$: the r equilibrium errors; α: the **adjustment coefficients** (loadings), how strongly each variable reacts to each error
  - only the product is unique: $\alpha\beta^\top = (\alpha Q)(\beta Q^{-\top})^\top$ for any invertible Q; we normalise β
- Testing for cointegration = testing the rank of Π (Section 5)

The Granger representation theorem

- **Theorem** Engle and Granger (1987): if $\mathbf{y}_t$ is $I(1)$ and cointegrated with rank r, then it has a VECM representation with $\Pi = \alpha\beta^\top$, and conversely
  - cointegration $\Leftrightarrow$ error correction: the equilibrium error must feed back into the changes
- **Consequences**
  - a VAR in differences omits $\alpha\beta^\top \mathbf{y}_{t-1}$: it is **misspecified** when the series are cointegrated
  - $\alpha \neq 0$: at least one variable adjusts, so there is Granger causality (Chapter 6) in at least one direction
  - the levels are driven by $n - r$ common stochastic trends; the effects of shocks on the levels do not die out
- A VAR in levels is not wrong either: it is consistent, but it ignores the restriction $\text{rank}(\Pi) = r$ and its tests are non-standard

2003: the Nobel Prize for cointegration



Clive Granger (1934–2009)

Photo: Olaf Storbeck (2008); CC BY-SA 2.0; Wikimedia Commons



The Sir Clive Granger Building, University of Nottingham

Photo: mattbuck (2012); CC BY-SA 3.0; Wikimedia Commons

- The 2003 prize: Granger for cointegration, shared with Robert Engle for ARCH (Chapter 5)
- Granger studied and began his career at Nottingham
- Nobel lecture: Granger (2004): "time series analysis, cointegration, and applications"

Worked example: a bivariate VECM

- $\Delta y_{1t} = -0.10\,(y_{1,t-1} - y_{2,t-1}) + u_{1t}, \quad \Delta y_{2t} = 0.10\,(y_{1,t-1} - y_{2,t-1}) + u_{2t}$
  - $\beta = (1,-1)^\top$, $\alpha = (-0.10, 0.10)^\top$; $\Pi = \alpha\beta^\top = \begin{pmatrix} -0.10 & 0.10 \\ 0.10 & -0.10 \end{pmatrix}$, $\det \Pi = 0$, rank 1
- If y_1 is above y_2: y_1 falls ($\alpha_1 < 0$) and y_2 rises ($\alpha_2 > 0$): both close the gap
- The gap $z_t = y_{1t} - y_{2t}$: $z_t = (1 + \beta^\top\alpha)\, z_{t-1} + (u_{1t} - u_{2t}) = 0.8\, z_{t-1} + \ldots$
  - a stationary AR(1); half-life $\ln 0.5 / \ln 0.8 = 3.1$ periods; the condition is $-2 < \beta^\top\alpha < 0$
- The average $(y_{1t} + y_{2t})/2$ changes by $(u_{1t} + u_{2t})/2$: a random walk, the common trend

Deterministic terms: five cases

Case	In the VECM	Implication for the data	`statsmodels`
1	no constant, no trend	no drift; relations through the origin	`n`
2	constant only inside $\beta^\top \mathbf{y}_{t-1}$	no drift; equilibrium errors with a non-zero mean (yields, spreads)	`ci`
3	unrestricted constant	linear trends in the levels, none in the relations (GDP, prices)	`co`
4	constant, and a trend inside $\beta^\top \mathbf{y}_{t-1}$	trends in the levels and in the relations	`coli`
5	unrestricted constant and trend	quadratic trends in the levels; rarely plausible	`colo`

- The critical values of the Johansen tests change with the case Johansen (1995); choose the case from the plot and from theory, before testing
- `coint_johansen` of `statsmodels` implements case 1 (`det_order=-1`) and case 3 (`det_order=0`), the case used in this chapter

Weak exogeneity

- If a row of α is zero, $\alpha_i = 0$, the variable y_i does **not** react to the equilibrium errors
  - y_i is **weakly exogenous** for β: the other variables do all the adjusting
  - y_i pushes the common trend; e.g. Euribor for Romanian interest rates (Seminar 7, B4)
- Test: a t-test on α_i (one cointegrating vector) or a likelihood-ratio (LR) test of $\alpha_i = 0$ Johansen (1991)
  - the statistic is asymptotically χ^2 with r degrees of freedom, once r is fixed
- Use: if x_t is weakly exogenous, a single-equation ECM for y_t (Section 3) loses no information about β

Recap: The VECM

- $\Delta \mathbf{y}_t = \alpha \beta^\top \mathbf{y}_{t-1} + \sum \Gamma_i \Delta \mathbf{y}_{t-i} + \mathbf{u}_t$: a VAR rewritten; $\text{rank}(\Pi) = r$
- Granger representation: cointegration $\Leftrightarrow$ error correction; a VAR in differences omits the long run
- β: equilibria; α: who adjusts; deterministic terms: five cases; $\alpha_i = 0$: weak exogeneity

The idea: reduced-rank regression



University of Copenhagen, where Søren Johansen and Katarina Juselius developed the method

Photo: Per Meistrup (2011); CC BY-SA 4.0; Wikimedia Commons

- Johansen (1988), Johansen (1991): maximum likelihood for the VECM with Gaussian errors
  - regress $\Delta \mathbf{y}_t$ and $\mathbf{y}_{t-1}$ on the lagged differences; keep the residuals R_{0t} and R_{1t}
  - find the combinations of R_{1t} most correlated with R_{0t}: the **canonical correlations**
- Their squares are the eigenvalues $1 > \hat{\lambda}_1 \geq \cdots \geq \hat{\lambda}_n \geq 0$
  - a large $\hat{\lambda}_i$: a stationary combination that predicts the changes; $\hat{\lambda}_i \approx 0$: a combination that is still $I(1)$
  - the eigenvectors give $\hat{\beta}$; then $\hat{\alpha}$ follows by OLS

Trace and maximum-eigenvalue statistics

- **Trace test**: H_0: rank $\leq r$, against rank $= n$; $\lambda_{\text{trace}}(r) = -T \sum_{i=r+1}^{n} \ln(1 - \hat{\lambda}_i)$
  - it sums the information in the $n - r$ smallest eigenvalues: if they are all close to 0, the statistic is small and H_0 is kept
- **Maximum-eigenvalue test**: H_0: rank $= r$, against rank $= r + 1$; $\lambda_{\max}(r) = -T \ln(1 - \hat{\lambda}_{r+1})$
  - it uses only the next eigenvalue, $\hat{\lambda}_{r+1}$; T: the number of observations
- **Sequential procedure**: test $r = 0$, then $r \leq 1$, ...; the rank is the first r whose H_0 is **not** rejected
  - the statistics are likelihood ratios, but their distributions are non-standard (functions of Brownian motions)
  - critical values depend on $n - r$ and on the deterministic case: Osterwald-Lenum (1992), MacKinnon, Haug and Michelis (1999)
- The two tests usually agree; if not, the trace test is preferred in most applied work

Worked example: from eigenvalues to statistics

- US yields at 1, 5 and 10 years, 801 months; effective sample $T = 801 - 2 - 1 = 798$: $\hat{\lambda} = (0.0405, 0.0229, 0.0039)$
  - $-\ln(1 - \hat{\lambda}_i)$: 0.0414; 0.0231; 0.0039
- $r = 0$: trace $= 798 \times (0.0414 + 0.0231 + 0.0039) = 54.6 > 29.80$: reject
  - max-eigenvalue: $798 \times 0.0414 = 33.0 > 21.13$: reject
- $r \leq 1$: trace $= 21.6 > 15.49$: reject; $r \leq 2$: trace $= 3.1 < 3.84$: do not reject
  - rank 2: two cointegrating vectors, one common trend
- Small eigenvalues still matter: they are multiplied by T

Johansen tests on three systems

System	T	k	$\lambda_{\text{trace}}(0)$	$\lambda_{\text{trace}}(1)$	$\lambda_{\text{trace}}(2)$	$\lambda_{\max}(0)$	$\lambda_{\max}(1)$	rank
US yields 1y, 5y, 10y	801	2	54.6	21.6	3.1	33.0	18.5	2
US output, consumption, investment	203	1	28.9	11.4	2.6	17.4	8.9	0
EUR/RON, EUR/HUF, EUR/PLN	255	1	24.2	7.9	2.7	16.3	5.2	0
5% critical value			29.80	15.49	3.84	21.13	14.26	

- Case 3 (unrestricted constant); k: lagged differences chosen by BIC on the VAR in levels; ranks by the trace test (the maximum-eigenvalue test gives the same ranks here)
- Yields: FRED, 1960–2026; output, consumption and investment: 100 ln, US 1959–2009; currencies: BNR, month-end, 2005–2026

Interpreting the three systems

- Yields: rank 2, so one common trend: the **level** of rates moves all maturities; the two spreads are stationary
  - the expectations hypothesis of the term structure Campbell and Shiller (1987) in its weak form
- Output, consumption and investment: trace 28.9 against 29.80: rank 0 at 5%, a near miss
  - balanced growth predicts rank 2 (stable shares of consumption and investment); with Engle–Granger, consumption and income were cointegrated ($p = 0.029$): low power, not proof of absence
- Currencies: trace 24.2: rank 0; three separate stochastic trends (Section 8)

The deterministic case changes the answer

Deterministic terms	$\lambda_{\text{trace}}(0)$	5%	$\lambda_{\text{trace}}(1)$	5%	$\lambda_{\text{trace}}(2)$	5%
none, case 1 (`det_order=-1`)	45.9	24.28	13.3	12.32	0.4	4.13
constant, case 3 (`det_order=0`)	54.6	29.80	21.6	15.49	3.1	3.84
constant and trend, case 5 (`det_order=1`)	60.0	35.01	23.7	18.40	4.8	3.84

- US yields at 1, 5 and 10 years: ranks 2, 2 and 3
- With a linear trend, the test says that every yield is stationary around a trend: not plausible for interest rates over 66 years
  - choose the deterministic terms from theory and from the plot, not from the result you like
- Other practical issues: the lag length k; size distortions in small samples (correction factor $(T - nk)/T$, Reinsel and Ahn (1992)); structural breaks

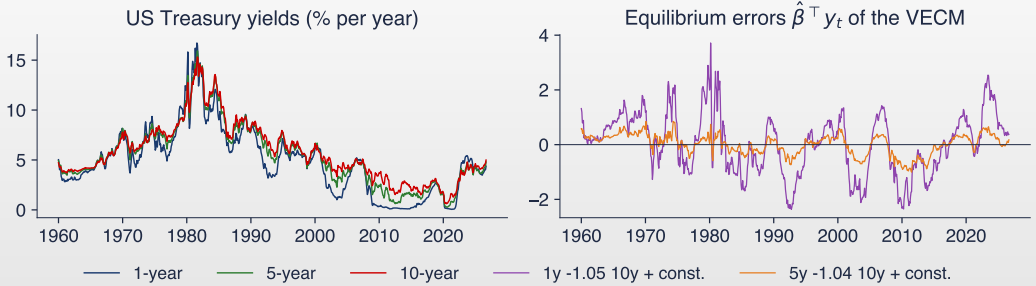
Case study: Johansen and Juselius (1990)

- **Question**: is there a stable demand for money in Denmark and Finland?
  - Johansen and Juselius (1990): real money, real income, interest rates (and inflation); quarterly data
- **Method**: the VECM by maximum likelihood; the rank by the trace and maximum-eigenvalue tests; then **tests of restrictions** on β and α
  - e.g. a unit income elasticity, $\beta = (1, -1, \cdot)$; weak exogeneity of income and interest rates
- **Legacy**: the template of the "cointegrated VAR" methodology Juselius (2006): test the rank, then test economic hypotheses as restrictions
  - one of the most cited papers in applied econometrics

Recap: The Johansen tests

- Eigenvalues of a reduced-rank regression; trace and maximum-eigenvalue statistics; sequential testing from $r = 0$
- Critical values depend on $n - r$ and on the deterministic case
- Yields: rank 2; macro aggregates: borderline; currencies: rank 0

The term-structure VECM



- US Treasury yields at 1, 5 and 10 years (FRED GS1, GS5, GS10), January 1960–September 2026; VECM with rank 2, 2 lagged differences, constant restricted to the relations (case 2)

TSA_ch7_johansen_vecm

Interpreting the cointegrating vectors

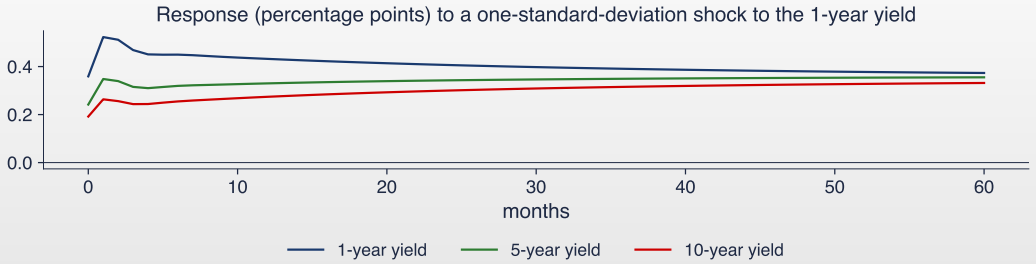
- Normalisation on the 1-year and 5-year yields: $\hat{\beta}_1^\top \mathbf{y}_t = i_t^{(1)} - 1.045\, i_t^{(10)} + 1.22$, $\hat{\beta}_2^\top \mathbf{y}_t = i_t^{(5)} - 1.044\, i_t^{(10)} + 0.59$
  - both slopes close to 1: the equilibria are almost the spreads $i^{(1)} - i^{(10)}$ and $i^{(5)} - i^{(10)}$, with mean term premia
- The equilibrium errors are stationary: ADF $p < 0.001$ and 0.008; they are large in the 1980s and around zero-rate periods
- A formal test of $\beta = (1, -1)$ for each spread is a likelihood-ratio test Johansen (1991); with super-consistent $\hat{\beta}$, values this close to 1 are informative

The adjustment coefficients

Equation	$\hat{\alpha}_{\cdot \mathbf{1}}$	t	$\hat{\alpha}_{\cdot \mathbf{2}}$	t
$\Delta i_t^{(\mathbf{1})}$ (1 year)	-0.021	-0.82	0.024	0.31
$\Delta i_t^{(\mathbf{5})}$ (5 years)	0.045	2.21	-0.122	-2.00
$\Delta i_t^{(\mathbf{10})}$ (10 years)	0.040	2.22	-0.077	-1.43

- The 1-year yield does not react to either error ($|t| < 1$): it behaves as **weakly exogenous**
  - the short end follows monetary policy; the 5- and 10-year yields do the adjusting ($|t|$ from 1.43 to 2.22)
- The equilibrium errors $\mathbf{z}_t = \hat{\beta}^\top \mathbf{y}_t$ follow $\mathbf{z}_t = (\boldsymbol{I} + \hat{\beta}^\top \hat{\alpha})\, \mathbf{z}_{t-\mathbf{1}} + \ldots$; eigenvalues 0.969 and 0.926
  - half-lives of about 22.1 and 9.0 months: slow error correction, typical of interest rates

Impulse responses of the VECM



⊡ Orthogonalised responses (Cholesky, ordering 1y, 5y, 10y; Chapter 6) to a one-standard-deviation shock to the 1-year yield

TSA_ch7_johansen_vecm

Interpreting the impulse responses

- On impact: +0.36, +0.24 and +0.19 percentage points for 1, 5 and 10 years
  - after 60 months: +0.37, +0.36 and +0.33: the effects do **not** die out and become almost equal
- A permanent shift of the common trend (the level), with stationary spreads: the signature of $n - r = 1$ common trend
- In a stationary VAR (Chapter 6) every response returns to zero; in a VECM the long-run responses are generally different from zero

Recap: β and α

- $\hat{\beta}$: normalise, read as equilibria (spreads, ratios), test economic restrictions
- $\hat{\alpha}$: who adjusts and how fast; a zero row means weak exogeneity
- Shocks have permanent effects on cointegrated levels, but not on the equilibrium errors

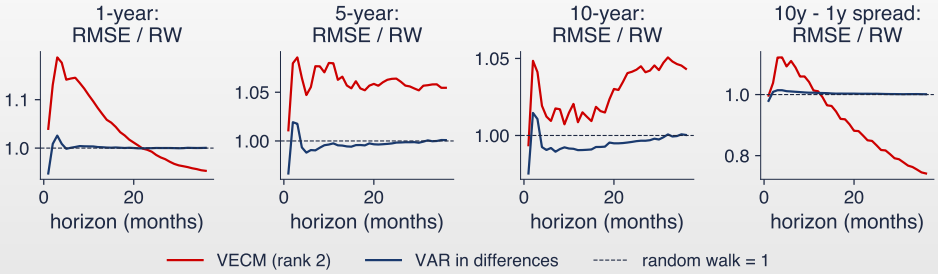
The theoretical arguments

- VECM forecasts keep the equilibria: as the horizon grows, $\hat{\beta}^\top \hat{\mathbf{y}}_{T+h}$ returns to its mean
  - a VAR in differences forgets the levels: its forecasts of the spreads stay where they are
- Engle and Yoo (1987): imposing cointegration improves long-horizon forecasts in simulated systems
- Christoffersen and Diebold (1998): the gains are concentrated in the **cointegrating combinations**
  - for the individual levels, the forecast error is dominated by the common trend, which no model can predict
  - estimation error in α, β and in the constants can make a VECM worse in practice
- Test it out of sample, with re-estimation at each forecast origin

The forecasting experiment

- Data: US yields at 1, 5 and 10 years, monthly, since 1960
- 135 forecast origins, every three months from January 1990 to July 2023; models re-estimated on the data up to each origin
  - VECM with rank 2 and a restricted constant; VAR in differences without constant, with the same lags; random walk (no change)
- Horizons of 1 to 36 months; RMSE (root mean squared error) of each model divided by that of the random walk
  - ratio below 1: better than the random walk; also for the 10-year minus 1-year spread

VECM against a VAR in differences, out of sample



- RMSE relative to the random walk by horizon; first three panels: the yields; last panel: the 10-year minus 1-year spread

TSA_ch7_forecasting

Interpreting the forecast comparison

- Levels: neither model beats the random walk clearly; the VECM is worse at short horizons (1 year at 6 months: 1.14; 5 years: 1.06)
  - the VAR in differences stays close to 1 at every horizon (1 year at 12 months: 1.00)
- Spread: the VECM gains at long horizons: 0.85 at 24 months and 0.74 at 36 months; the VAR in differences: 1.00
  - exactly the pattern of Christoffersen and Diebold (1998): cointegration helps forecast the equilibrium errors, not the trend
- Lesson: use a VECM when the object of interest is the relation (a spread, a ratio, a pair); for the levels alone, a random walk is hard to beat

Recap: Forecasting

- The VECM pulls the forecasts of the equilibrium errors back to their means; the VAR in differences does not
- Out of sample: gains for the spread at long horizons, none for the levels
- Always compare with the random walk, with re-estimation and without look-ahead

Purchasing power parity

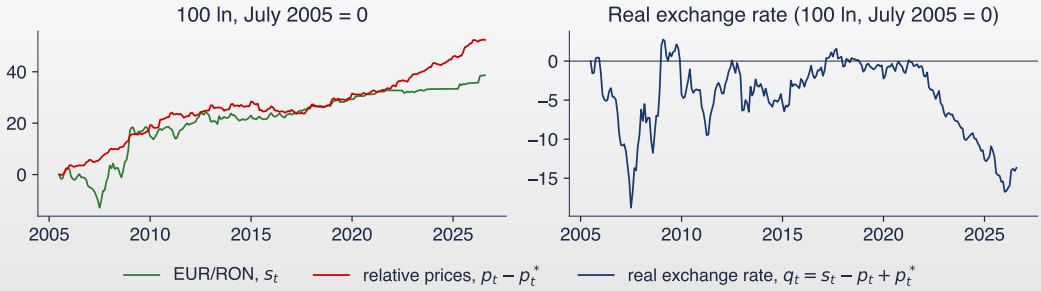


The National Bank of Romania, which publishes the reference rates

Photo: Ștefan Jurcă (2015); CC BY 2.0; Wikimedia Commons

- **Relative PPP** (purchasing power parity): the exchange rate offsets inflation differences
  - $s_t = 100 \ln(\text{EUR/RON})$; p_t, p_t^*: $100 \ln$ of the HICP (Harmonised Index of Consumer Prices) of Romania and of the euro area
  - PPP holds in the long run if the real exchange rate $q_t = s_t - p_t + p_t^*$ is stationary: cointegration with $\beta = (1, -1, 1)$
- Reasons why it may fail
  - Balassa (1964): fast productivity growth in tradables raises prices of non-tradables in a catching-up economy: a trend real appreciation
  - even where PPP holds, deviations decay slowly; the survey Taylor and Taylor (2004) reports half-lives of several years

PPP for the leu



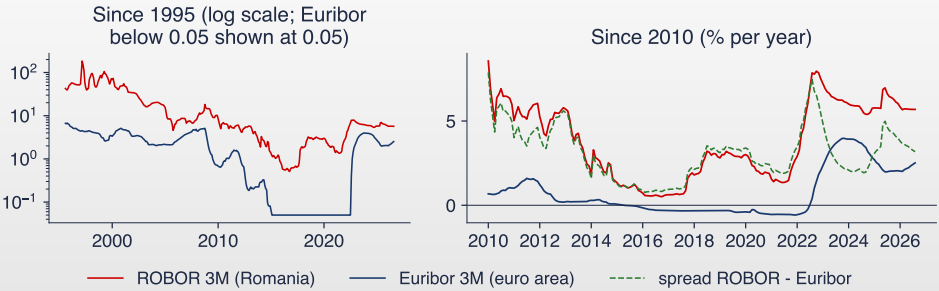
- Monthly, July 2005–August 2026: EUR/RON (monthly mean of the BNR reference rate), HICP of Romania and of the euro area (Eurostat, 2015 = 100)

TSA_ch7_parity_conditions

Interpreting PPP for the leu

- Since 2005 the leu lost 38.7 log points against the euro, while Romanian prices rose 52.4 log points more than euro-area prices
  - the real exchange rate changed by -13.7 log points: a real appreciation of the leu, concentrated after 2021
- ADF on q_t: p $= 0.24$; Engle–Granger of s_t on $p_t - p_t^*$: slope 0.89, p $= 0.34$; Johansen on (s_t, p_t, p_t^*): rank 0
- No evidence of PPP over 21 years, consistent with Balassa–Samuelson convergence; 21 years may also be too short for half-lives of several years

ROBOR and Euribor



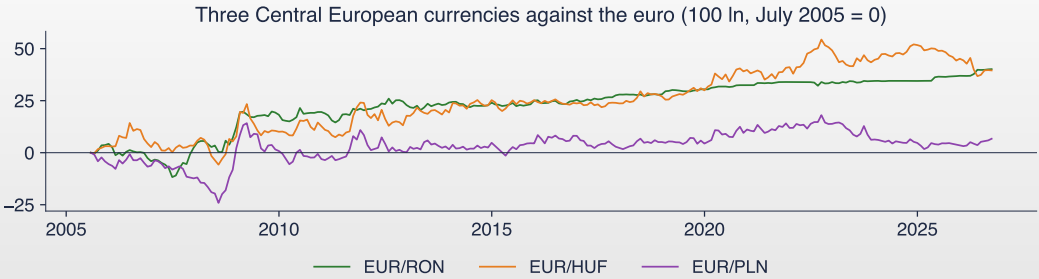
- Three-month money market rates (Eurostat), August 1995–August 2026: ROBOR 3M for Romania and Euribor 3M for the euro area

TSA_ch7_parity_conditions

Interpreting ROBOR and Euribor

- Full sample: Engle–Granger p = 0.52, Phillips–Ouliaris p = 0.010: the disinflation of the 1990s dominates and the tests disagree
- Since 2010: p = 0.047 (Engle–Granger) and 0.029 (Phillips–Ouliaris), slope 1.19, mean spread 3.11 percentage points
  - ECM for ROBOR: $\hat{\gamma} = -0.020$ ($t = -1.18$), half-life about 34 months: weak and slow adjustment
- Romanian monetary policy follows its own inflation target; the link to Euribor is loose (Seminar 7, B4, adds the Romanian 10-year yield)

Three Central European currencies



- EUR/RON, EUR/HUF and EUR/PLN, cross rates from the BNR reference rates, month-end, 100 ln

TSA_ch7_parity_conditions

Interpreting the three currencies

- Since July 2005: EUR/RON +40.1, EUR/HUF +39.5, EUR/PLN +6.7 log points; ADF does not reject a unit root at 5% for any rate (p = 0.55; 0.75; 0.08, EUR/PLN borderline at 10%)
- Johansen: trace 24.2 for $r = 0$ (critical value 29.80): no evidence of cointegration, also since 2012 (trace 12.7)
  - a common anchor (the euro) and similar shocks do not create a common trend: each central bank and each inflation path adds its own
- The right model for these rates is a VAR in differences (Chapter 6); Seminar 7 (B3) repeats the analysis

Recap: Economic examples

- No evidence of PPP for the leu over 2005–2026 (a unit root in q_t is not rejected): real appreciation of a converging economy
- ROBOR and Euribor: a loose, slow link since 2010; the sample decides the verdict
- Theory proposes β; the tests, the sample and the deterministic terms decide whether the data agree

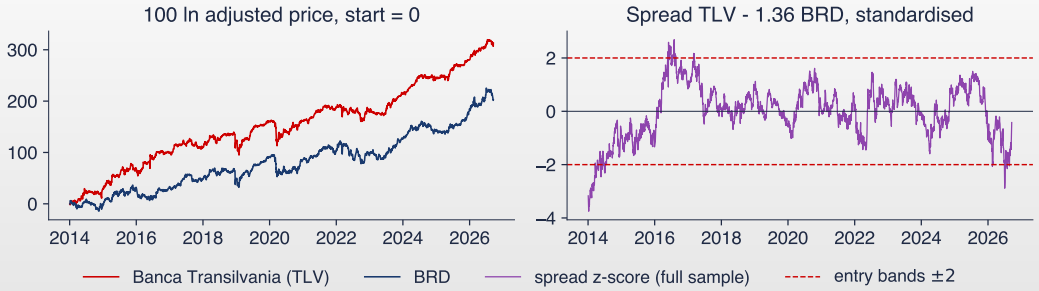
The idea of pairs trading

- Two shares exposed to the same shocks (two banks, two oil companies) may be cointegrated: their spread mean-reverts
  - when the spread is unusually wide, sell the expensive share and buy the cheap one; close when the spread is back
- A **relative-value** trade: the market direction cancels out; the profit comes from the error correction
  - Gatev, Goetzmann and Rouwenhorst (2006): the distance method on US shares, profitable over 1962–2002
  - Do and Faff (2010): the profits declined in later years
- A good test of a time-series idea: it must work **out of sample**, **after costs**, and on pairs chosen **without looking ahead**

A rule without look-ahead

- **Formation** (the previous 252 trading days): test every pair with Engle–Granger; keep the pairs with p < 0.05
  - $P_{A,t}$, $P_{B,t}$: the prices of shares A and B; r_A, r_B: their daily log returns
  - hedge ratio $\hat{b}$ by OLS of $\ln P_A$ on $\ln P_B$; mean $\bar{s}$ and standard deviation $\hat{\sigma}_s$ of the spread $s_t = \ln P_{A,t} - \hat{a} - \hat{b} \ln P_{B,t}$
- **Trading** (the next 126 days), with the formation parameters fixed: $z_t = (s_t - \bar{s})/\hat{\sigma}_s$
  - $z_t > 2$: short A, long $\hat{b}$ of B; $z_t < -2$: the opposite; close at $z_t = 0$ or at the end of the window
  - daily return per unit of gross exposure: $\pm(r_A - \hat{b}\, r_B)/(1 + |\hat{b}|)$; equal capital per selected pair
- **Costs**: 0.20% (Bucharest) and 0.05% (US) of the gross exposure traded, at each opening and closing

Banca Transilvania and BRD



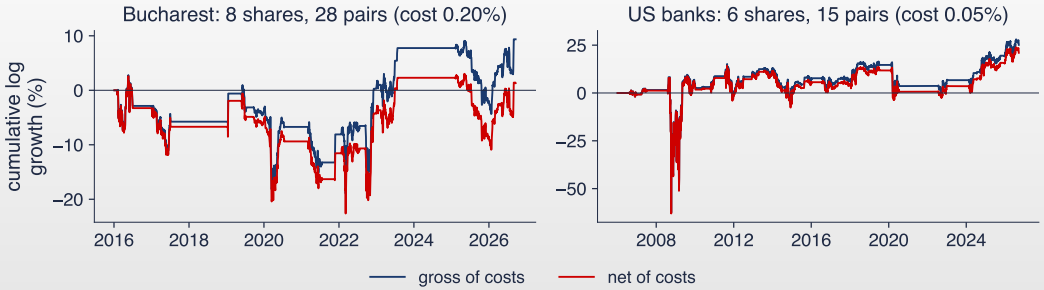
- Adjusted daily prices since 2014; right: the spread of the full-sample regression, standardised, with the entry bands

TSA_ch7_pairs_trading

Interpreting the bank pair

- Full sample: $\hat{b} = 1.36$; the spread is an AR(1) with $\hat{\rho} = 0.990$: half-life about 69 trading days
  - only 5.0% of the days lie outside ± 2: few trading opportunities, each lasting months
- Trading this picture with full-sample parameters gives 3.32% per year (Sharpe ratio 0.44), after costs
  - Sharpe ratio: the mean annual return divided by the annual standard deviation of the returns
  - an illusion: $\hat{b}$, $\bar{s}$ and $\hat{\sigma}_s$ use the future, and the pair was chosen because it looks cointegrated since 2014
- With the rolling rule, the pair passes the formation test in none of its 24 windows: one year of data is too short to detect such slow error correction

Pairs trading out of sample



- Left: 8 liquid BVB shares (TLV, BRD, SNP, TGN, TEL, EL, SNG, SNN), since 2015; right: 6 US banks (JPM, BAC, C, WFC, GS, MS), since 2005

TSA_ch7_pairs_trading

Interpreting the backtest

- Bucharest: 22 windows, pairs found in 14, 73 trades, 62% of trades profitable, but the mean return per trade is -0.26%
  - gross 1.17% per year (Sharpe 0.15); net 0.41% (Sharpe 0.05); break-even cost 0.31%
- US banks: gross 2.77% (Sharpe 0.16), net 2.57% (Sharpe 0.14), 80 trades
  - with a deep loss in the autumn of 2008: cointegration can break exactly when it matters
- Many small wins and a few large losses: the typical profile of a mean-reversion strategy

Limitations of pairs trading

- **Multiple testing**: with 28 pairs and a 5% test, about 1.4 pairs pass by chance in each window
- **Look-ahead**: choosing pairs, start dates or thresholds after seeing the whole sample
- **Breaks**: mergers, recapitalisations, regulation and crises change β; the spread may never come back
- **Market frictions**: short selling is limited on the BVB, borrowing shares costs a fee, bid-ask spreads are wide for small caps
- **Slow error correction**: half-lives of months mean capital tied up for months, and a long wait in a losing position

Recap: Pairs trading

- Pairs trading bets on error correction in the spread of two cointegrated shares
- In sample it looks attractive; out of sample, after costs, the profits on the BVB are close to zero
- Report the selection, the costs, the number of trades and the worst episode, not only the Sharpe ratio

Possible contribution of AI

- **Code**: a first draft of a script that runs unit-root tests, Engle–Granger, Phillips–Ouliaris and Johansen on many series and tabulates the verdicts
- **Explanation**: a second explanation of the Granger representation theorem, or of a `VECM` output table
- **Exploration**: candidate pairs on the BVB, or parity conditions for several Central European countries
- Example prompt
  - `Write Python code that loads monthly US Treasury yields GS1, GS5 and GS10 from FRED, runs the Johansen trace test with a constant and BIC lags, fits a VECM with the chosen rank using statsmodels, and prints beta normalised on the first variables, alpha with t-statistics and the half-lives of the equilibrium errors.`

Checks you must run

- The critical values: Engle–Granger (MacKinnon) for residuals, not Dickey–Fuller; Johansen for the right deterministic case
- The direction of the null: "no cointegration" in Engle–Granger; "rank $\leq r$" in the trace test
- That every series is $I(1)$ before testing, and that a high correlation of levels is not reported as cointegration
- The sign of the adjustment coefficients and the half-lives, in the right time unit
- Backtests: no look-ahead in pair selection, costs included, the number of pairs tested reported
- Every cited reference: it must exist; check the DOI

Key takeaways

- Cointegrated $I(1)$ series share common stochastic trends; a combination $\beta^\top \mathbf{y}_t$ is stationary
- Engle–Granger: OLS in levels, then ADF on the residuals with MacKinnon critical values; check with Phillips–Ouliaris
- ECM: changes respond to the lagged equilibrium error; γ gives the speed and the half-life
- VECM: $\Pi = \alpha\beta^\top$; Granger representation: cointegration $\Leftrightarrow$ error correction; Johansen tests choose r
- Cointegration helps forecast relations (spreads), not levels; applications must be judged out of sample and after costs

Key formulas

Quantity	Formula
Cointegration	$\mathbf{y}_t \sim I(1), \quad \beta^\top \mathbf{y}_t \sim I(0)$
Engle–Granger	$y_t = a + bx_t + u_t; \quad \Delta\hat{u}_t = \gamma\hat{u}_{t-\mathbf{1}} + \sum_j \delta_j \Delta\hat{u}_{t-j} + e_t, \quad H_\mathbf{0}\colon \gamma = 0$
ECM	$\Delta y_t = c + \gamma\,\hat{u}_{t-\mathbf{1}} + \delta_\mathbf{0}\Delta x_t + \cdots + \varepsilon_t, \quad h_{\mathbf{1/2}} = \ln 0.5/\ln(1+\gamma)$
VECM	$\Delta\mathbf{y}_t = \alpha\beta^\top \mathbf{y}_{t-\mathbf{1}} + \sum_{i=\mathbf{1}}^{p-\mathbf{1}} \Gamma_i \Delta\mathbf{y}_{t-i} + \mathbf{u}_t, \quad \Pi = \sum_i A_i - I, \quad \Gamma_i = -\sum_{j>i} A_j$
Johansen	$\lambda_{\mathbf{trace}}(r) = -T\sum_{i>r}\ln(1-\hat{\lambda}_i), \quad \lambda_{\max}(r) = -T\ln(1-\hat{\lambda}_{r+\mathbf{1}})$
Equilibrium errors	$\mathbf{z}_t = \beta^\top \mathbf{y}_t\colon \quad \mathbf{z}_t = (I + \beta^\top\alpha)\mathbf{z}_{t-\mathbf{1}} + \ldots$

Self-assessment

- **Question**: Engle–Granger with two variables gives $\tau = -3.1$ for $T = 300$. Do you reject "no cointegration" at 5%?
  - **Answer**: no: $-3.1 > -3.36$; with the Dickey–Fuller value (-2.87) you would wrongly reject
- **Question**: an ECM gives $\hat{\gamma} = -0.1$ on monthly data. What is the half-life?
  - **Answer**: $\ln 0.5 / \ln 0.9 \approx 6.6$ months
- **Question**: four $I(1)$ variables have cointegration rank 3. How many common trends drive them?
  - **Answer**: $n - r = 1$
- Next: Chapter 8, long memory: series between $I(0)$ and $I(1)$, fractional differencing and ARFIMA

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